IC104 Introductory Quantitative Techniques for Business and Finance

Lecture 1. Introduction to Algebra

Dr Nikolaos Antypas

ICMA Centre
Henley Business School
University of Reading

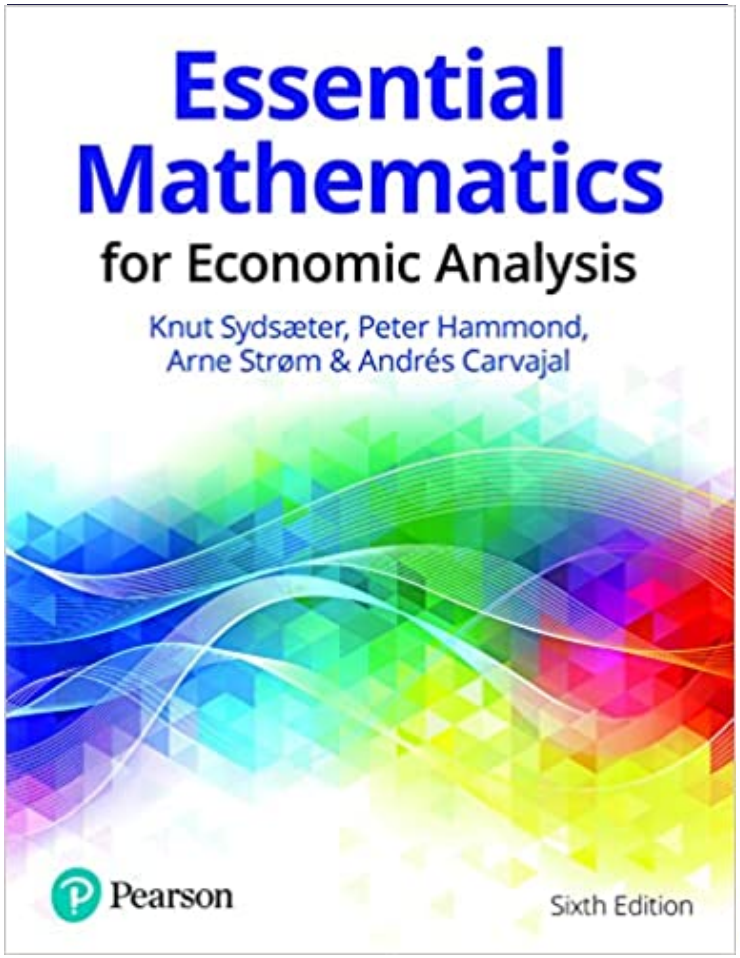
About the Course

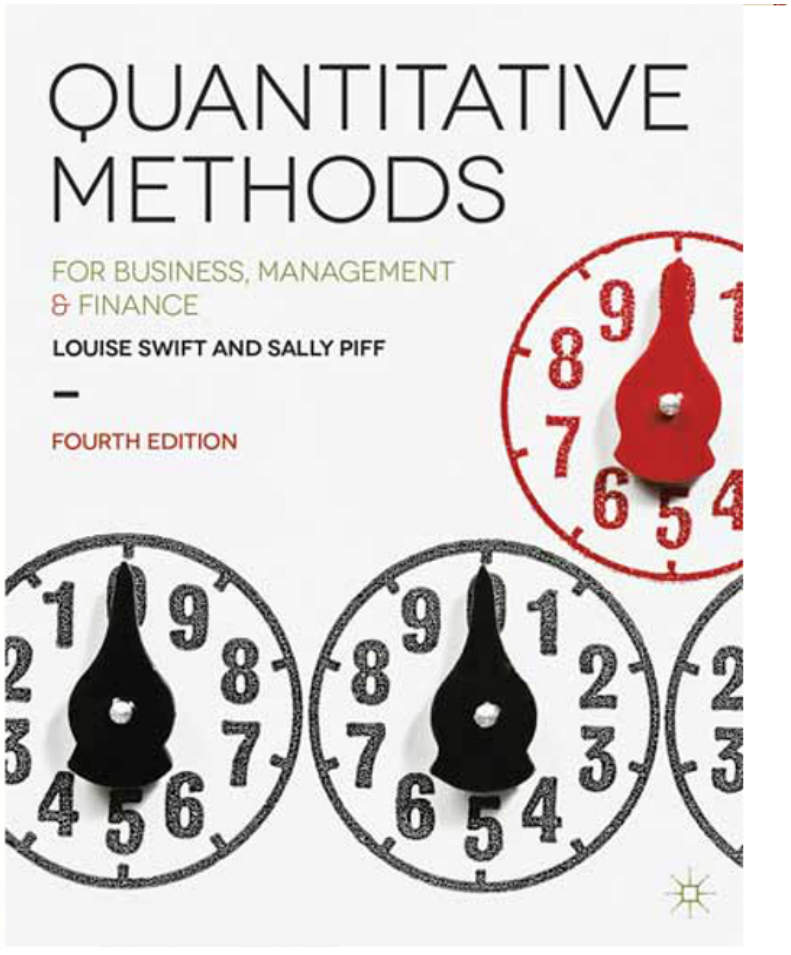
- Lecture: in person - days may vary
- Office hours in Semester 1: Tuesday & Friday, 13.00-14.00
- Any variations will be posted to Blackboard. At other times by
- appointment.
- E-mail: n.antypas@henley.ac.uk, Office G60 in ICMA Centre

Teaching Assistants

- Your first port of call for help.
- They are always available in their Office hours - they're here to help run the tutorials, and to address your questions.
- Details of their Office Hours will become available on Blackboard under ”Module Information > Teaching Team”.
- If many students request to see them, it may help to email in advance and reserve a time slot.

Course Books





Evaluation

- Evaluation breakdown:
  - Project 25%
  - Final Exam 75%
- The project will be given near the end of Teaching, to be submitted by Assessment Week 1.
- More details later on in the course.
- Final exam in Assessment Period.
- A minimum module mark of 40% for a pass.

Course Website

- See the Blackboard website - look for the course on the Enrolments tab: www.bb.reading.ac.uk
- This website holds the main resource for the course:
  - Lecture notes
  - Tutorial questions and solutions
  - Contact details and Office Hours, with any updates
  - Supplementary materials
  - Announcements
- It will greatly help your progress if you work through the questions before the tutorials. The tutorial questions will be worked through in the tutorials, and the answers will be made available on Blackboard after the relevant tutorial.
- Please check your timetables for details of your tutorials. Tutorials begin in Week 7. This is during week commencing October 12th.

Course Road Map

- Mathematics
  - Week 1: Introduction to Algebra
  - Week 2-3: Differentiation, Partial Differentiation, Single Variable Optimization and Integration
  - Week 3-4: Topics in Financial Economics and Matrix Introduction
  - Week 4-5: Matrix Algebra
- Statistics
  - Week 6: Data Description and Probability I
  - Week 7: Probability II
  - Week 8: Estimation
  - Week 9: Testing Hypotheses
  - Week 10: Correlation and Regression

Advice for the course

- Warning: the Mathematics section of the course may be difficult, depending on how much maths you've done before.
- Many future courses rely upon the mathematical foundations built up here, so the work you put in now will pay dividends later on when you come to its applications.
- Mathematics is the language of modern finance and economics
- Come to lectures prepared
- Ask for help

Lecture Plan

- Algebra Basics (Sydsaeter & Hammond Chapters 1 & 2)
- Linear and Equations (Sydsaeter & Hammond Chapter 3)

- Exponential and Logarithmic Functions (Sydsaeter & Hammond Chapter 4.9 & 4.10)

Integer Powers

- If a is any number and n is any natural number, then a^n is defined by

$$a^n = \underbrace{a \cdot a \cdot \ldots \cdot a}_{n \text{ factors}}$$

a^n is called the **nth power** of a
a is the **base**, n is the **exponent**
$a^0 = 1$ for $a \neq 0$
$a^{-n} = \frac{1}{a^n}$

- Properties of Powers

$$a^r \cdot a^s = a^{r+s} \qquad (a^r)^s = a^{rs}$$

Integer Powers

- Simply the following:

  1) $x^p x^{2p} =$

  2) $t^s \div t^{s-1} =$

  3) $a^2 b^3 a^{-1} b^5 =$

  4) $\dfrac{t^p \cdot t^{q-1}}{t^r \cdot t^{s-1}} =$

- Answers:

  1) $x^p x^{2p} = x^{p+2p} = x^{3p}$

  2) $t^s \div t^{s-1} = t^{s-(s-1)} = t^{s-s+1} = t$

  3) $a^2 b^3 a^{-1} b^5 = a^2 a^{-1} b^3 b^5 = a^{2-1} b^{3+5} = ab^8$

  4) $\dfrac{t^p \cdot t^{q-1}}{t^r \cdot t^{s-1}} = \dfrac{t^{p+q-1}}{t^{r+s-1}} = t^{p+q-1-(r+s-1)} = t^{p+q-r-s}$

Application of Integer Powers: Compound Interest

- If you deposit £1,000 in a bank account paying 8% interest at the end of each year:

  After one year: $1000 + 1000 \cdot 8\% = 1000 \cdot (1 + 8\%)$
  After two years: $1000 \cdot (1 + 8\%) \cdot (1 + 8\%) = 1000 \cdot (1 + 8\%)^2$
  After three years:
  $1000 \cdot (1 + 8\%) \cdot (1 + 8\%) \cdot (1 + 8\%) = 1000 \cdot (1 + 8\%)^3$
  After t years: $1000 \cdot (1 + 8\%)^t$

- If the original amount is £K and the interest rate is $p\%$ per year, the value after t years will be:

$$K(1 + p\%)^t$$

Rules of Algebra

If a, b and c are arbitrary numbers, then:

(a) $a + b = b + a$

(b) $(a + b) + c = a + (b + c)$

(c) $a + 0 = a$

(d) $a + (-a) = 0$

(e) $ab = ba$

(f) $(ab)c = a(bc)$

(g) $1 \cdot a = a$

(h) $a^0 = 1$ for $a \neq 0$

(i) $(-a)b = a(-b) = -(ab)$

(j) $(-a)(-b) = ab$

(k) $a(b + c) = ab + ac$

(l) $(a + b)c = ac + bc$

Quadratic Identities:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

Examples

Expand each of the following

- $(3x + 2y)^2$
- $(1 - 2z)^2$
- $(4p + 5q)(4p - 5q)$
- $(r + 1)^3$

Answers:

- $(3x + 2y)^2 = (3x)^2 + 2(3x)(2y) + (2y)^2 = 9x^2 + 12xy + 4y^2$
- $(1 - 2z)^2 = 1 - 2 \cdot 1 \cdot 2 \cdot z + (2z)^2 = 1 - 4z + 4z^2$
- $(4p + 5q)(4p - 5q) = (4p)^2 - (5q)^2 = 16p^2 - 25q^2$
- $(r + 1)^3 = [(r + 1)(r + 1)](r + 1) = (r^2 + 2r + 1)(r + 1) =$
  $(r^2 + 2r + 1)r + (r^2 + 2r + 1)1 =$
  $r^3 + 2r^2 + r + r^2 + 2r + 1 = r^3 + 3r^2 + 3r + 1$

Fractions

Rule

1. $\frac{a \cdot c}{b \cdot c} = \frac{a}{b} \quad b \neq 0 \text{ and } c \neq 0$
2. $\frac{-a}{-b} = \frac{(-a) \cdot (-1)}{(-b) \cdot (-1)} = \frac{a}{b}$
3. $-\frac{a}{b} = (-1)\frac{a}{b} = \frac{(-1)a}{b} = \frac{-a}{b}$
4. $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$
5. $\frac{a}{b} + \frac{c}{d} = \frac{a \cdot d + b \cdot c}{b \cdot d}$
6. $a + \frac{b}{c} = \frac{a \cdot c + b}{c}$
7. $a \cdot \frac{b}{c} = \frac{a \cdot b}{c}$
8. $\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$
9. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{a \cdot d}{b \cdot c}$

Example

1. $\frac{21}{15} = \frac{7 \cdot 3}{5 \cdot 3} = \frac{7}{5}$
2. $\frac{-5}{-6} = \frac{5}{6}$
3. $-\frac{13}{15} = (-1)\frac{13}{15} = \frac{(-1)13}{15} = \frac{-13}{15}$
4. $\frac{5}{3} + \frac{13}{3} = \frac{18}{3} = 6$
5. $\frac{3}{5} + \frac{1}{6} = \frac{3 \cdot 6 + 5 \cdot 1}{5 \cdot 6} = \frac{23}{30}$
6. $5 + \frac{3}{5} = \frac{5 \cdot 5 + 3}{5} = \frac{28}{5}$
7. $7 \cdot \frac{3}{5} = \frac{21}{5}$
8. $\frac{4}{7} \cdot \frac{5}{8} = \frac{4 \cdot 5}{7 \cdot 8} = \frac{4 \cdot 5}{7 \cdot 2 \cdot 4} = \frac{5}{14}$
9. $\frac{3}{8} \div \frac{6}{14} = \frac{3}{8} \cdot \frac{14}{6} = \frac{3 \cdot 2 \cdot 7}{2 \cdot 2 \cdot 2 \cdot 2 \cdot 3} = \frac{7}{8}$

Fractional Powers

If a and b are non-negative numbers, then:

- $\sqrt{ab} = \sqrt{a}\sqrt{b}$
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, $b \neq 0$
- the nth root of a: $a^{\frac{1}{n}} = \sqrt[n]{a}$

Fractional Powers

Simplify the following expressions so that the answers contain only positive exponents:

- $\dfrac{a^{\frac{3}{8}}}{a^{\frac{1}{8}}}$
- $(x^{\frac{1}{2}}x^{\frac{3}{2}}x^{-\frac{2}{3}})^{\frac{3}{4}}$
- $(\dfrac{10p^{-1}q^{\frac{2}{3}}}{80p^2q^{-\frac{7}{3}}})^{-\frac{2}{3}}$

Answers: $\dfrac{a^{\frac{3}{8}}}{a^{\frac{1}{8}}} = a^{\frac{3}{8}-\frac{1}{8}} = a^{\frac{2}{8}} = a^{\frac{1}{4}} = \sqrt[4]{a}$

$(x^{\frac{1}{2}}x^{\frac{3}{2}}x^{-\frac{2}{3}})^{\frac{3}{4}} = (x^{\frac{1}{2}+\frac{3}{2}-\frac{2}{3}})^{\frac{3}{4}} = (x^{\frac{4}{3}})^{\frac{3}{4}} = x$

$(\dfrac{10p^{-1}q^{\frac{2}{3}}}{80p^2q^{-\frac{7}{3}}})^{-\frac{2}{3}} = (8^{-1}p^{-1-2}q^{\frac{2}{3}-(-\frac{7}{3})})^{-\frac{2}{3}} = 8^{\frac{2}{3}}p^2q^{-2} = 4\dfrac{p^2}{q^2}$

Inequalities

- $a > 0$ and $b > 0$ implies that $a + b > 0$ and $a \cdot b > 0$
- If $a > b$, then $a + c > b + c$ for all c
- $a > b$ and $b > c \Rightarrow a > c$
- $a > b$ and $c > 0 \Rightarrow ac > bc$
- $a > b$ and $c < 0 \Rightarrow ac < bc$
- $a > b$ and $c > d \Rightarrow a + c > b + d$

Intervals and Absolute Values

Notation	Name	The interval consists of all x satisfying:
(a, b)	The **open** interval from a to b	$a < x < b$
$[a, b]$	The **closed** interval from a to b	$a \leq x \leq b$
$(a, b]$	The **half-open** interval from a to b	$a < x \leq b$
$[a, b)$	The **half-open** interval from a to b	$a \leq x < b$
$[a, \infty)$	Unbounded interval	$x \geq a$
$(-\infty, b)$	Unbounded interval	$x < b$

The **absolute value** of a is denoted by $|a|$, and

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Summation

- Summation Notation (the sigma operator)

$$N_1 + N_2 + N_3 + \cdots + N_n = \sum_{i=1}^{n} N_i$$

i is called the index of summation.

- Rules for Sums
- $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$
- $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$, where c is a constant
- $\sum_{i=1}^{n} k = nk$, where k is a constant

Summation

- Common Errors:

$$\sum_{i=1}^{n} p_i q_i \neq \sum_{i=1}^{n} p_i \sum_{i=1}^{n} q_i,$$

$$\sum_{i=1}^{n} (x_i^2) \neq (\sum_{i=1}^{n} x_i)^2$$

- Double Sums:

$$\sum_{i=1}^{m} a_{i1} + \sum_{i=1}^{m} a_{i2} + \cdots + \sum_{i=1}^{m} a_{in} = \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}$$

Summation

Useful Formulas:

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = \frac{1}{2} n(n+1)$$

$$\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

$$\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + \cdots + n^3 = (\frac{1}{2} n(n+1))^2 = (\sum_{i=1}^{n} i)^2$$

Summation

Useful sigma equalities: $\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$

Proof:

$$\sum(x_i - \bar{x})^2 = \sum(x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$
$$= \sum x_i^2 - 2\sum x_i\bar{x} + \sum \bar{x}^2$$
$$= \sum x_i^2 - 2n\bar{x}\bar{x} + n\bar{x}^2$$
$$= \sum x_i^2 - n\bar{x}^2$$

$\bar{x}$ is the arithmetic mean of the set of numbers $\{x_1, x_2, ..., x_n\}$

In other words, $\bar{x} = \frac{\sum x_i}{n}$ and $n\bar{x} = \sum x_i$

Terms and Concepts

- An equation is any mathematical expression that contains an "equals" sign.
- An equation is said to be **linear** if NONE of the variables that appear in it is raised to any power other than power 1.
- For example:

$$y - 3x + 2 = 0 \text{ or } \frac{3}{4}x = -3$$

- Examples of non-linear equations:

$$y^2 - 3x + 2 = 0 \text{ or } \frac{1}{x} + 2 = 0$$

- An equation may contain one or more **unknowns** or **variables** and also some given or known values, called constants, coefficients or parameters.

$y - 3x = -2$: two variables (x and y) and two coefficients (3 and 2)

$y - ax + b = 0$: two variables (x and y) and two parameters (a and b)

Independent and Dependent Variables

- In general, a function between two variables can be written as

  $y = f(x)$, where f denotes the relationship between the two variables, x and y.

- The variable that appears on the right-hand side of the equation (here, x) is conventionally called the **independent** variable, because it is free to take any value we choose to assign to it.

- The variable on the left-hand side of the equation (here, y) is generally called the **dependent** variable. This is because the value of y depends on the value x.

- We say that "y is a function of x".

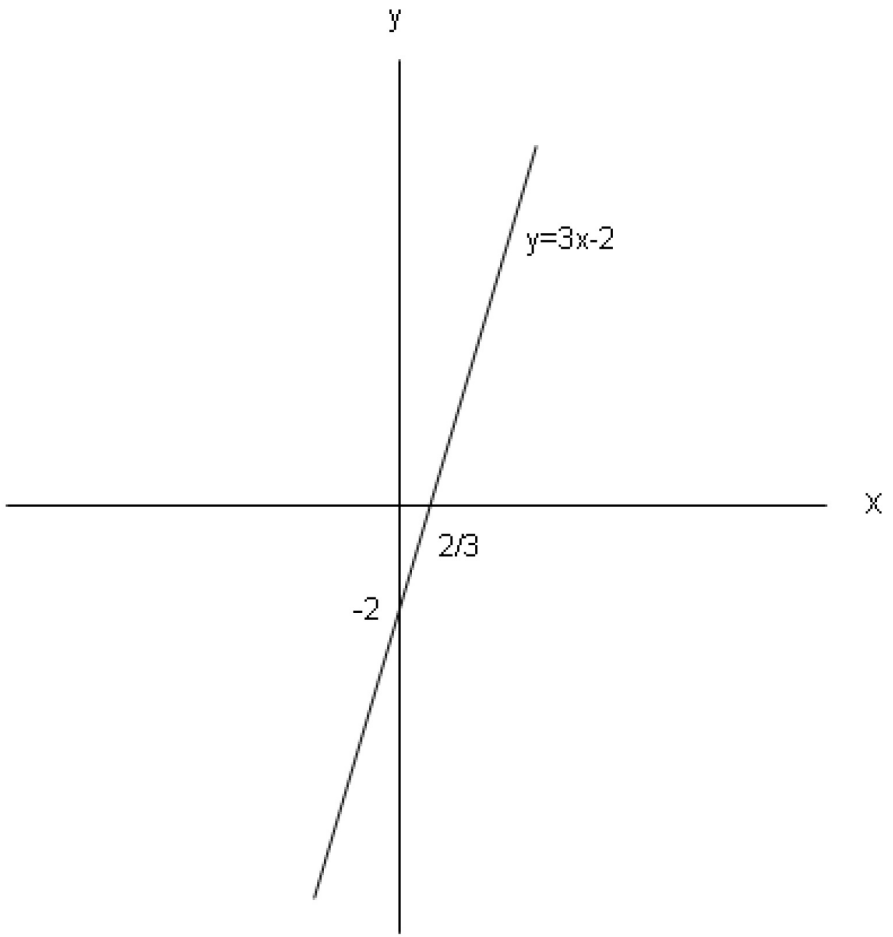
Graphs of Linear Functions

- Consider the equation (or function): $y = 3x - 2$.
- We can show the relationship between x and y by plotting the equation on a graph. First, find at least two points (or coordinates) that satisfy the equation.

x	y=3x-2
-2	y = 3 (-2) - 2 = -8
-1	y = 3 (-1) - 2 = -5
0	y = 3 (0) - 2 = -2
1	y = 3 (1) - 2 = 1
2	y = 3 (2) - 2 = 4

- Note that (-2, -8) and (0, -2) are pairs of values for x and y that satisfy the equation y = 3x - 2. They are also points on the graph of y = 3x - 2, and are called the **coordinates** of those points. By convention, the first number in the brackets is the **x**-coordinate, while the second number is the **y**-coordinate.

Graph of $y = 3x - 2$



The Slope and the Intercept

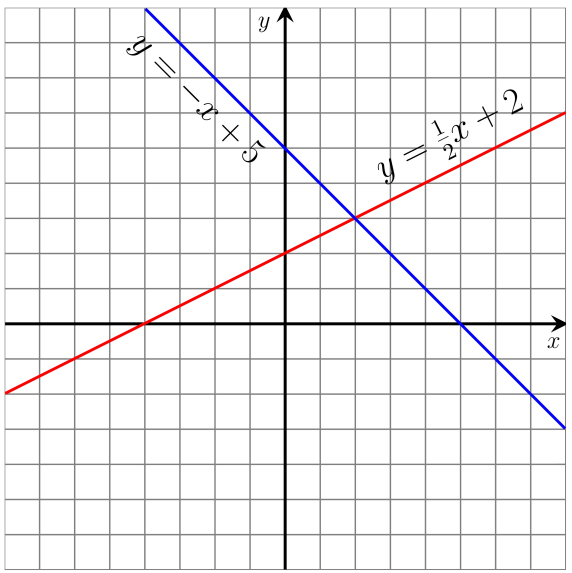
- In general, a linear function $y = ax + b$ (where a and b are any two constants):

  has a graph that is a straight line.

  has a **slope** of a: This is the measure of how much y increases when we increase x by one unit.

  cuts the y-axis at $y = b$ (the **y-intercept**), and x-axis at $x = -\frac{b}{a}$ (the x-intercept).

  slopes upwards if $a > 0$ (the slope is positive), and slopes downwards if $a < 0$ (the slope is negative). If $a = 0$, the graph is a horizontal line.



Implicit Linear Functions

- Sometimes a linear function is presented as $4x - 2y + 6 = 0$

  This is an **implicit** function: both x and y appear on the same side of the equation.

- If the implicit function is linear, it is easy to transform it into the familiar linear form, where y is on the left-hand side of the equation and x is on the right-hand side: $y = 2x + 3$

- In general, given an implicit linear function: $Ax + By + C = 0$

  Rearranging it into the **explicit** form: $y = -\frac{A}{B}x - \frac{C}{B}$

  The slope is $-\frac{A}{B}$, and the intercept with the y-axis is $-\frac{C}{B}$.

Simultaneous Linear Equations or System of Equations

- Consider if these two equations are true at the same time:

$$y = 4x$$
$$y = 2x + 6$$

- They are called **simultaneous** linear equations.
- To solve them, equate them, using the fact that the Left Hand Sides of both are the same.
- We have: $4x = 2x + 6$
- Hence, $x = 3$. Substitute $x = 3$ into one of the equations to find y.
- For instance, $y = 2(3) + 6 = 12$. So $x = 3$, $y = 12$ is the solution.

Methods of finding the solution

- Consider now the simultaneous equations

$$y = 3x - 5$$

$$\frac{4 + y}{2} = x$$

- We cannot immediately proceed as before, since neither the left-hand nor the right-hand sides are equal to one another.
- Method 1. We manipulate the equations so that either the left-hand or the right-hand sides are equal.

  For the equations above, take $\frac{4 + y}{2} = x$ and solve it for y.

  Multiply both sides by 2 to get $4 + y = 2x$, then subtract 4 from both sides to get $y = 2x - 4$.

  Now equate this with the first equation: $y = 3x - 5 = 2x - 4$.

  $x = 1$, so that $y = -2$.

Methods of finding the solution

- Consider now the simultaneous equations

$$y = 3x - 5$$
$$\frac{4 + y}{2} = x$$

- Method 2. The substitution method. Here we replace one variable (y) in one of the equations with the y in the other equation.

Substitute $y = 3x - 5$ into the second equation:

$$\frac{4 + (3x - 5)}{2} = x$$
$$\frac{(3x - 1)}{2} = x$$
$$3x - 1 = 2x$$
$$x = 1, \text{ so } y = -2.$$

A Model of Demand and Supply

- Demand and Supply are the basic building blocks for economic analysis.
- A model of demand and supply describes how prices affect the behavior of both consumers and producers.
- In general, the quantity demanded of a good sold in a market depends not only on its own price but also on other factors, such as the prices of other goods and consumers' income.
- For example, if the price of Pepsi increases while the price of Coca-Cola remains constant, some consumers may switch and consume more Coca-Cola if they feel that the two products are close substitutes (i.e. they are very similar products).

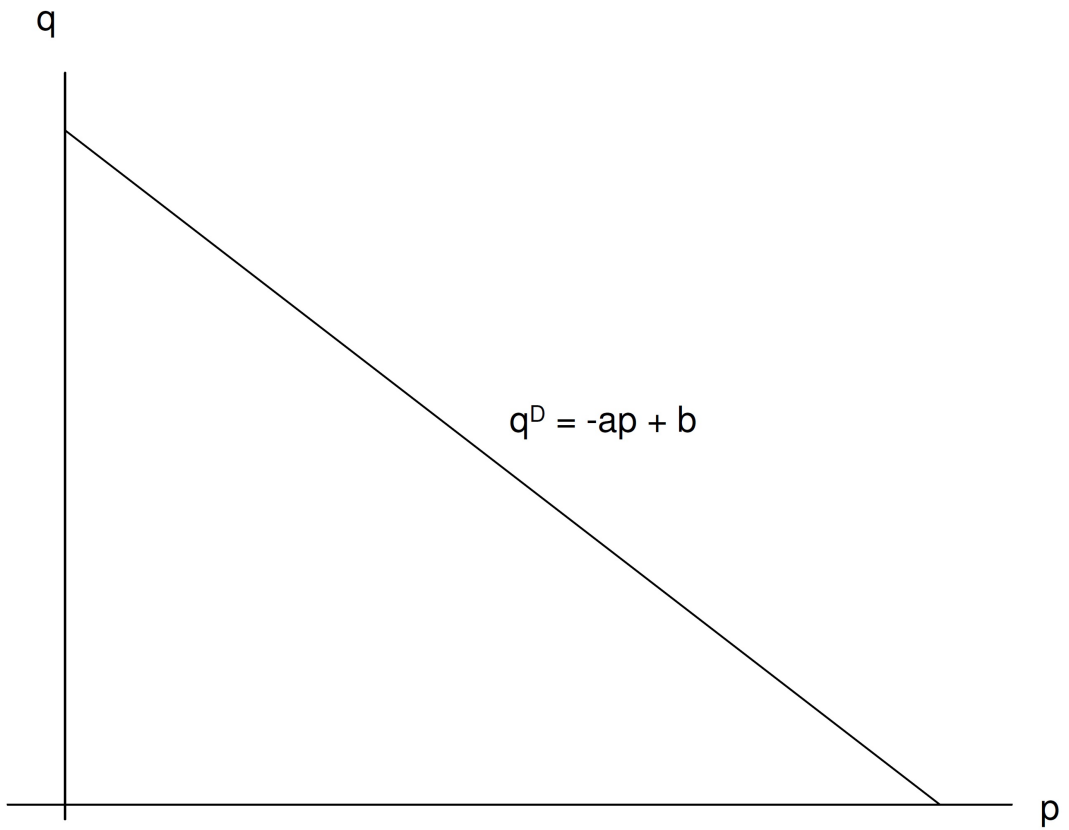
Demand

- The concept of a demand function is based on consumer theory: the lower the price of the good is, the more the consumer should want to buy it (taking the other factors as given).
- Adding up all the individual demands gives us the market demand:

  $q^D = -ap + b$, where p is the total market price and a and b are positive parameters.

- The demand curve is negatively sloped: The higher the price of the good, the lower the quantity customers demand of that good – **Law of Demand**.
- The quantity demanded (q^D) is the dependent variable, and we assume that, for each individual, the market price (p) is given so that she/he chooses how much to buy at that price.

Demand curve



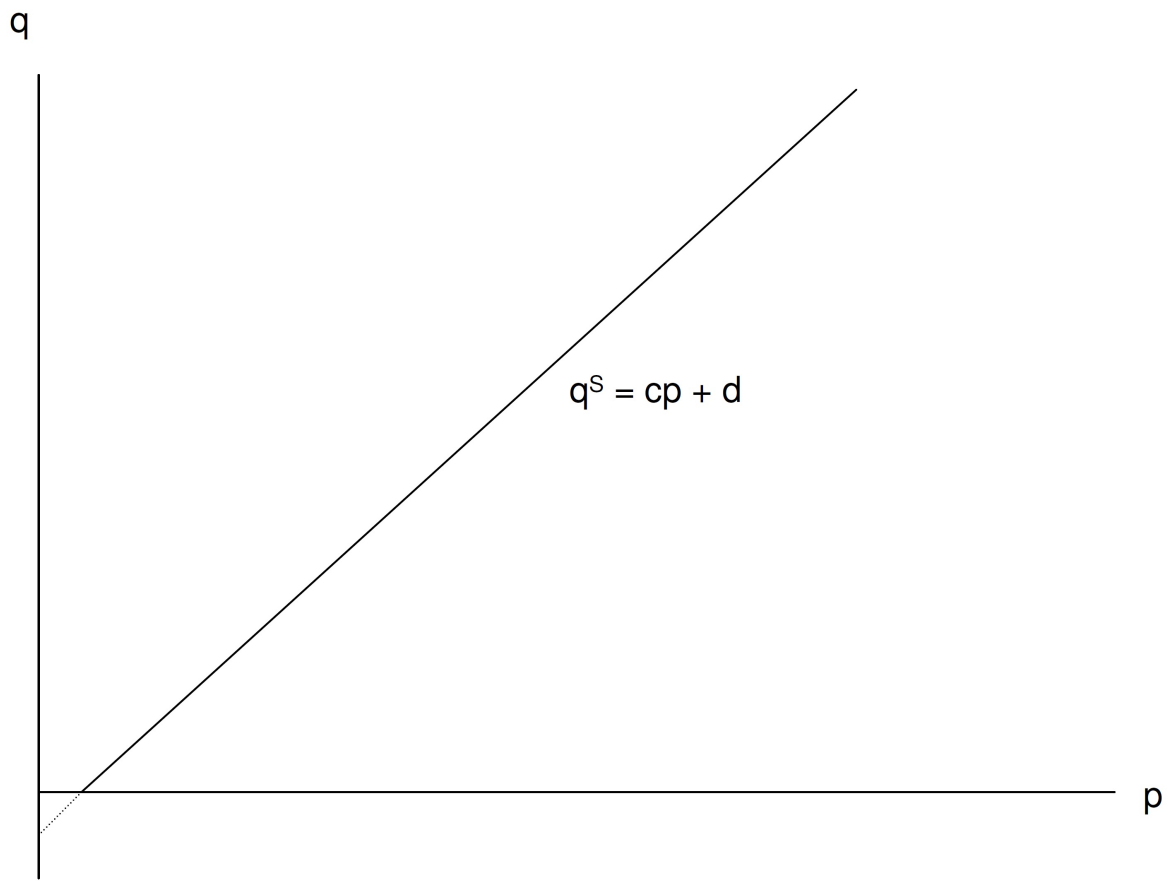
Supply

- The concept of the supply function is based on the theory of firm: The higher is the market price of the good, the more firms will choose to supply that good.
- Combining all firms supply gives us the market supply:

  $q^S = cp + d$, where $c > 0$ and d is zero or negative (with zero price firms, wish to supply nothing).

- The supply curve is positively sloped: The higher is the price of the good the higher is the quantity supplied of that good – **Law of Supply**.
- Quantity supplied (q^S) is the dependent variable and price (p) is assumed to be given.

Supply Curve



Market Equilibrium

- An **equilibrium** is a situation where none of the agents have an incentive to change anything.
- The market is 'in equilibrium' when the quantity that the consumers are willing to buy equals the quantity the producers are willing to supply:

$$q^D = q^S \text{ (equilibrium condition)}$$

- If $q^D > q^S$, then some consumers would like to buy more than is supplied in the market by firms.
- If $q^D < q^S$, then some producers would like to supply more than is demanded in the market by consumers.

Example

- Suppose the demand for books is specified as the following relationship between quantity (Q) and price (P) :

  $$q^D = 40 - 2p$$

- The supply of books is specified as another relationship:

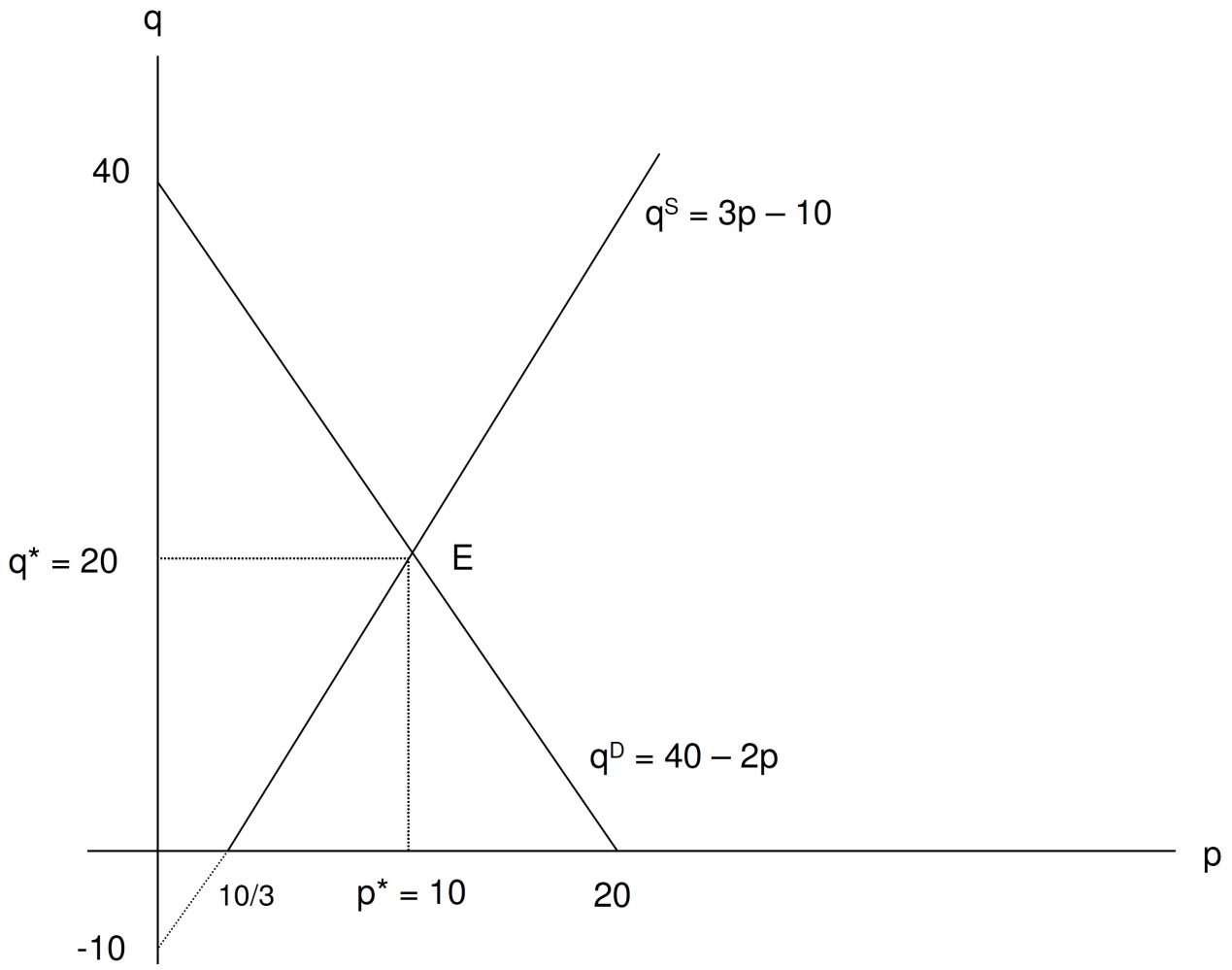
  $$q^S = 3p - 10$$

- The equilibrium condition is: $q^D = q^S$.
- This is a system of three linear equations and three unknowns (q^D, q^S, p). To solve for the equilibrium, we set:

  $40 - 2p = 3p - 10$, and solve for p: $p^\star = 10$

- Plug this into one of the equations above to get $q^\star = 3(10) - 10 = 20$.

Market Equilibrium



Solving for the Market Equilibrium: A general case

- Consider the three equations in a demand and supply model:

$$q^D = -ap + b$$
$$q^S = cp + d$$
$$q^D = q^S$$

- Set $-ap + b = cp + d \Rightarrow p^\star = \dfrac{b-d}{a+c}$
- $q^D = -ap + b, \Rightarrow q^\star = \dfrac{ad+bc}{a+c}$

Quadratic Equations

- A **quadratic equation** is a mathematical expression that contains an "equals sign" and an unknown variable that is raised to the power **2**:

$$ax^2 + bx + c = 0 \ (a \neq 0)$$

- If $b^2 - 4ac \geq 0$, the two solutions, called 'roots of the equation,' are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- This is a key formula - memorize it .

Quadratic Equations

- Quadratic formula: $ax^2 + bx + c = 0 \ (a \neq 0)$
- If $b^2 - 4ac \geq 0$, the two solutions are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

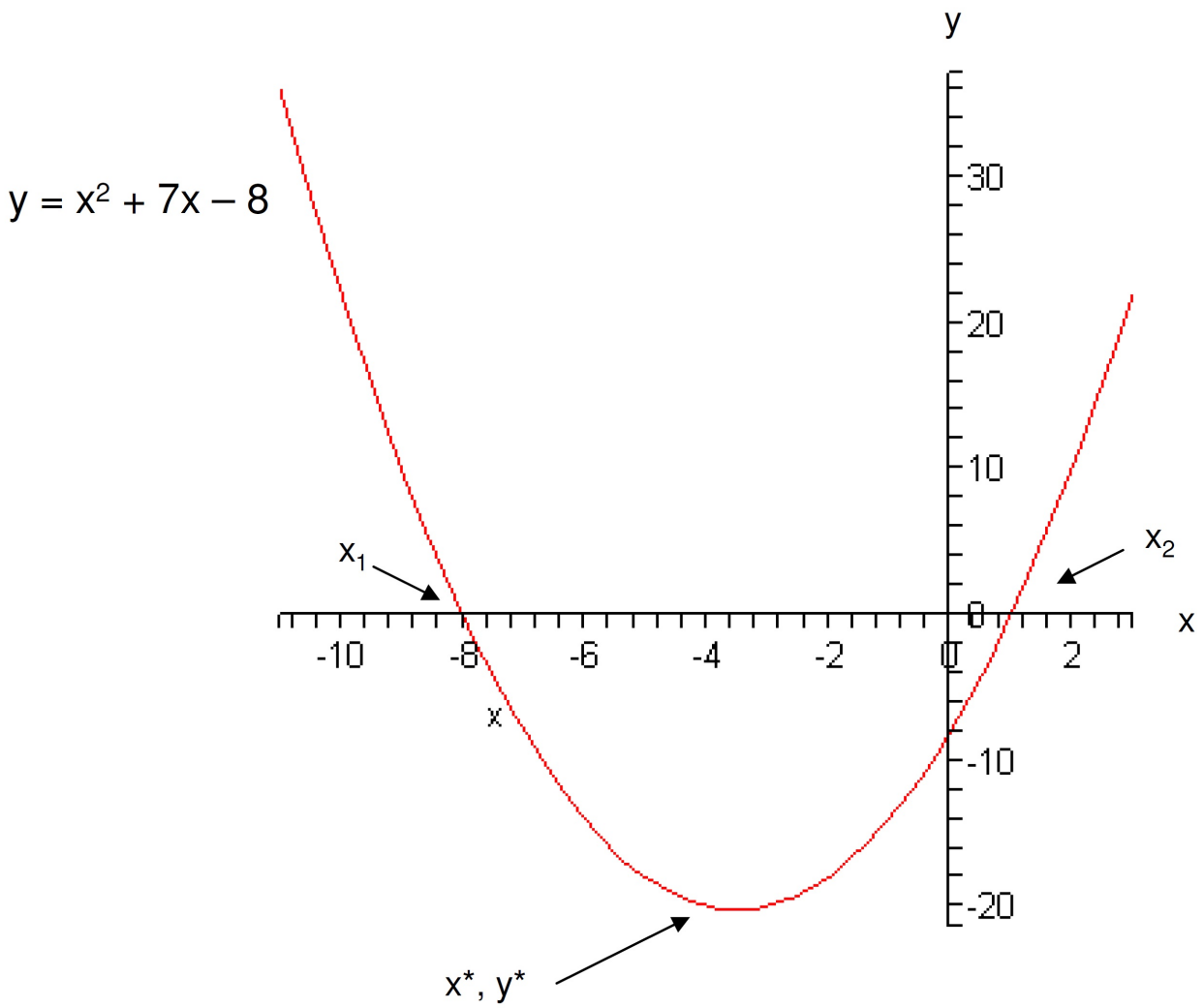
- Example: $x^2 + 7x - 8 = 0$

$a = 1,\ b = +7,\ c = -8$

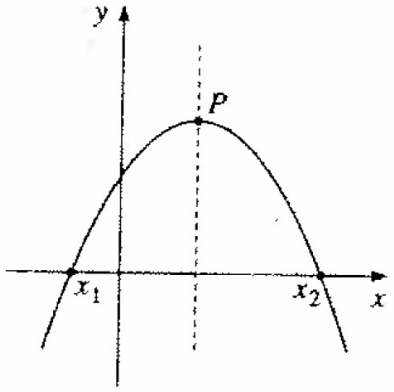
$$x = \frac{-(+7) \pm \sqrt{(+7)^2 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1} = \frac{-7 \pm \sqrt{49 + 32}}{2} = \frac{-7 \pm 9}{2}$$

$x_1 = -8$, and $x_2 = 1$

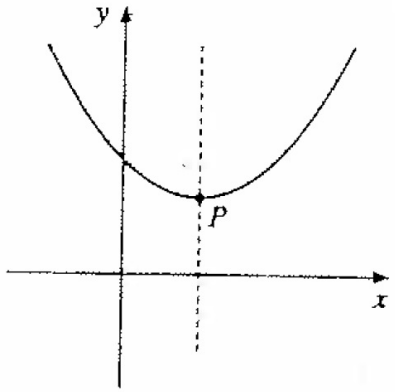
Graphing Quadratic Functions (Parabola)



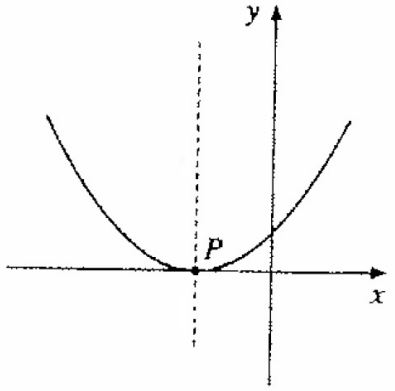
Graphing Quadratic Functions: General Cases



(a) $a < 0, b^2 > 4ac$



(b) $a > 0, b^2 < 4ac$



(c) $a > 0, b^2 = 4ac$

- We can decompose a general quadratic function into components:

$$f(x) = ax^2 + bx + c = a(x + \frac{b}{2a})^2 - \frac{b^2 - 4ac}{4a}$$

- If $a > 0$, then $f(x)$ has a **minimum** at $x = -\frac{b}{2a}$
- If $a < 0$, then $f(x)$ has a **maximum** at $x = -\frac{b}{2a}$

Example

- The price P per unit obtained by a firm in producing and selling Q units is $P = 102 - 2Q$, and the total cost of producing and selling all these Q units is $C = 2Q + \frac{1}{2}Q^2$. Then the profit is:

$$\pi(Q) = PQ - C = (102 - 2Q)Q - (2Q + \frac{1}{2}Q^2) = 100Q - \frac{5}{2}Q^2$$

- Find the value of Q which maximizes profits, and the corresponding maximal profit.
- In this case, $a = -\frac{5}{2} < 0$, so:

$$Q = Q^\star = -\frac{b}{2a} = -\frac{100}{2(-\frac{5}{2})} = 20 \text{ with}$$

$$\pi^\star = \pi(Q^\star) = 100*20 - \frac{5}{2} * 20^2 = 1000$$

Exponential Functions

- A quantity that increases (or decreases) by a fixed ratio per unit of time is said to increase (or decrease) exponentially.

$$f(x) = Aa^x \text{ where } a \text{ is positive}$$

- If $f(x) = Aa^x$, then $f(x+1) = Aa^{x+1} = Aa^x \cdot a = af(x)$

  If $a > 1$, then f is increasing with increasing x

  If $a < 1$, then f is decreasing with increasing x

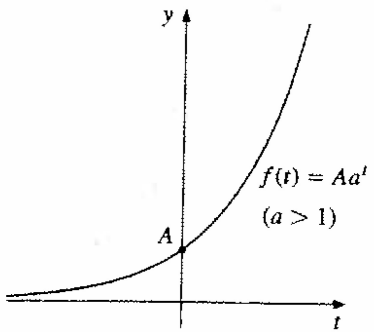


Figure 1 Graph of $f(t) = Aa^t$ $(a > 1)$

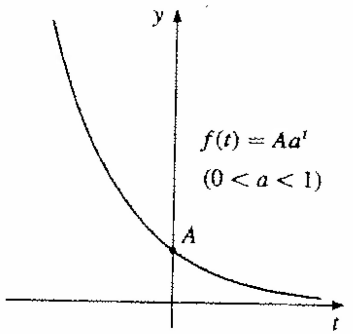


Figure 2 Graph of $f(t) = Aa^t$ $(0 < a < 1)$

- Note that there is a difference between: $f(x) = a^x$ and $g(x) = x^a$

Compound Interest

- In 1624, Peter Minuit bought Manhattan Island from the American Indians for $ 24.
- If the Indians invested the $24 at 10%, in 1624, how much would it be have been worth in 2012 (388 years)?

$$24(1.1)^{388} = \$275,786,550,708,197,000$$

- 2012 US GDP: $15.8 trillion (12 zeros)

$$\$275,786,550,708,197,000/\$20,500,000,000,000 = 13,453$$

- i.e 13,453 times US GDP.

Depreciation

- Each year the value of most assets such as cars or furniture decreases, or *depreciates*.
- Assume that a car, which at time $t = 0$, has value of P_0, depreciates at the rate of 20% each year over a 5 year period. What is its value $A(t)$ at time t, for $t = 1, 2, 3, 4, 5$?
- Answers: After 1 year, its value is $P_0 - P_0 \cdot 20\% = P_0 \cdot (0.8)^1$. Thus, each year it depreciates by the factor 0.8, so after t years, its value is $A(t) = P_0(0.8)^t$

Natural Exponential Function

- Given the exponential function $f(x) = Aa^x$ $(A = 1,\ a = e)$, we have the natural exponential function:

$$f(x) = e^x \text{ where } e = 2.718281828459045\cdots$$

- The exponential function e^x can be characterized in a variety of equivalent ways

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$e^x = \int_1^y \frac{dt}{t}$$

$$e^x = \lim_{n\to\infty} (1 + \frac{x}{n})^n$$

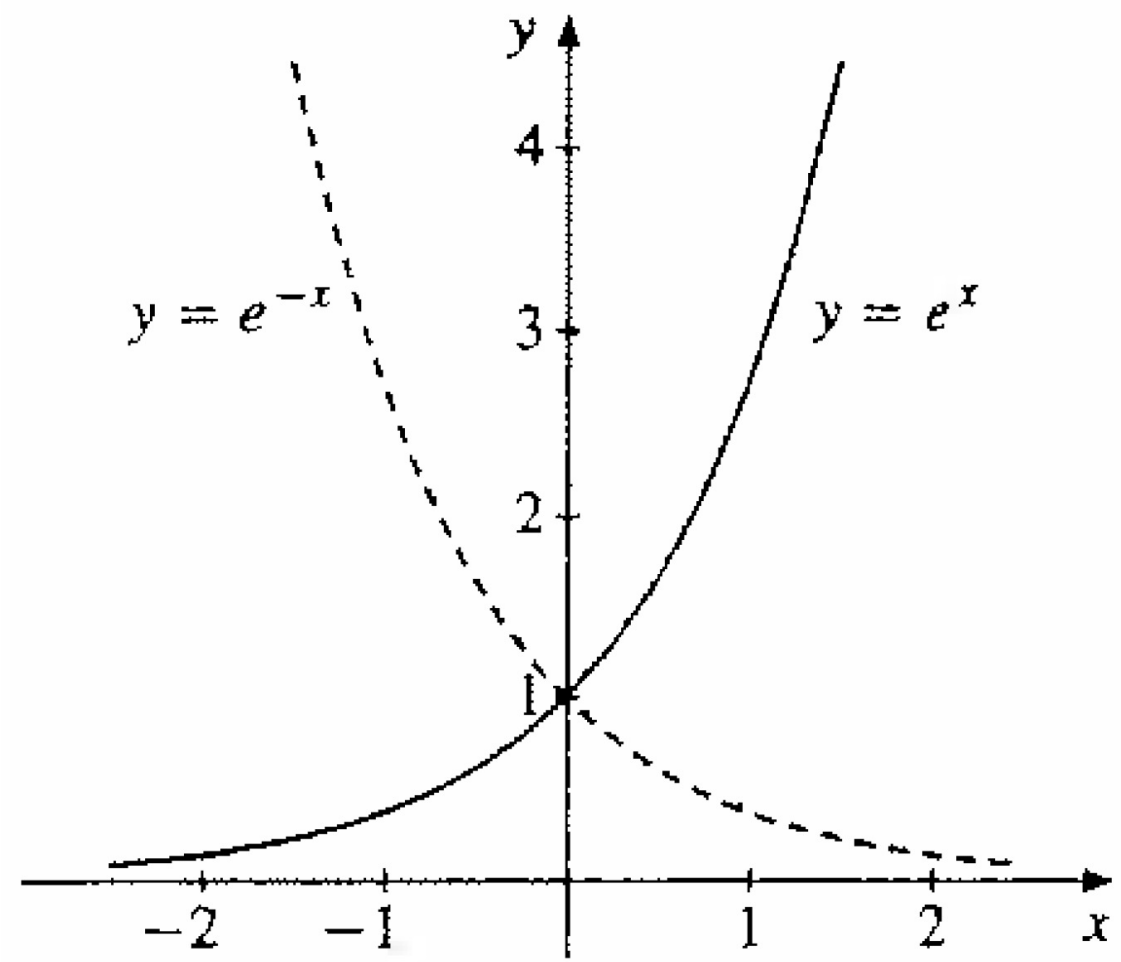
- Properties of the natural exponential functions:

$$e^s e^t = e^{s+t},\ \frac{e^s}{e^t} = e^{s-t},\ (e^s)^t = e^{st}$$

Natural Exponential Function

- What is so important about e ?
- If $f(x) = e^x$, then the slope of the graph of $f(x)$ at any point is exactly equal to e^x .
- This makes the Differentiation (see next week) of e^x much easier.

Graphs of Natural Exponential Functions



Logarithmic Functions

- Consider the equations: (1) $e^x = 4$, (2) $5e^{-3x} = 16$, (3) $Aae^{-ax} = k$
- In all three equations, the unknown x occurs as an exponent. To solve for x, we introduce the following definition:

  If $e^u = a$, we call u the **natural logarithm** of a

  $u = \ln a$, and $e^{\ln a} = a$ for $a > 0$

- Natural logarithm function properties:

  $\ln 1 = 0$, $\ln e = 1$, $\ln(\frac{1}{e}) = -1$

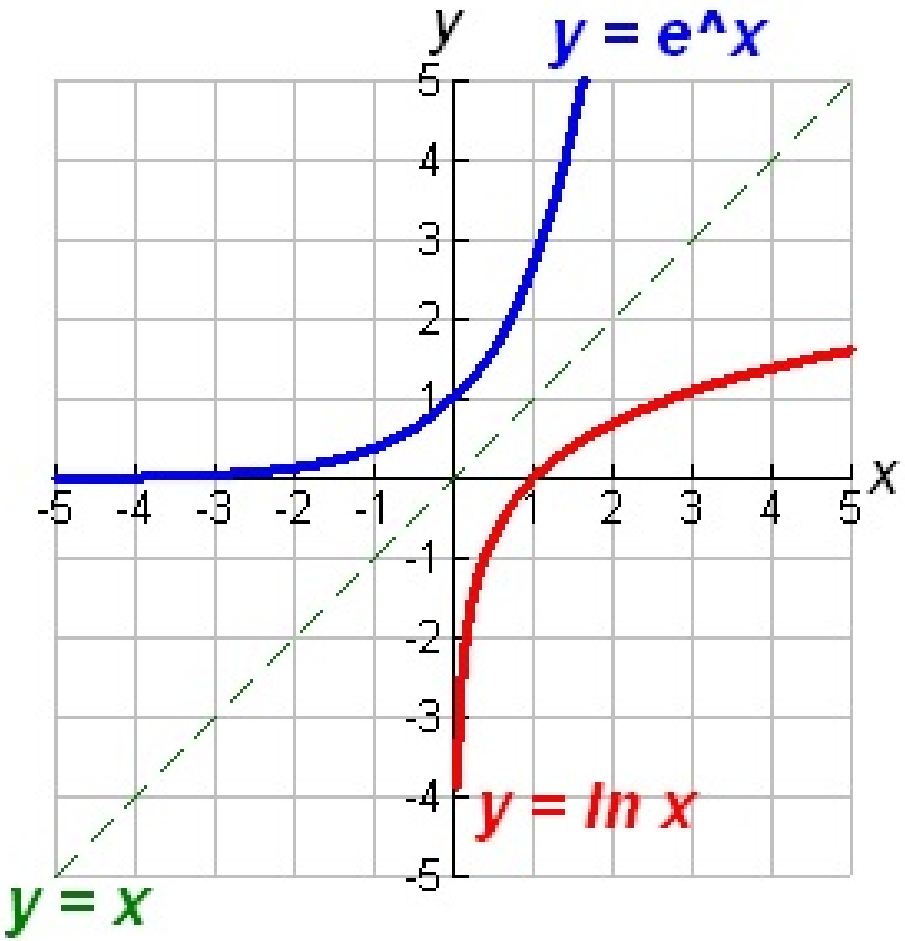
  $\ln xy = \ln x + \ln y$, for $x > 0$ and $y > 0$

  $\ln \frac{x}{y} = \ln x - \ln y$, for $x > 0$ and $y > 0$

  $\ln e^p = p$

  $x = e^{\ln x}$ for $x > 0$

The Function $g(x) = \ln x$



Logarithms with Bases other than e

- If $a^u = x$, then we call u the logarithm of x to base a and write: $u = \log_a x$
- $\log_a x$ is then defined for every positive number x by $a^{\log_a x} = x$
- Take ln on each side, we obtain $\log_a x \cdot \ln a = \ln x$, so that

$$\log_a x = \frac{1}{\ln a} \ln x$$

- Properties:

$$\log_a xy = \log_a x + \log_a y, \text{ for } x > 0 \text{ and } y > 0$$

$$\log_a \frac{x}{y} = \log_a x - \log_a y, \text{ for } x > 0 \text{ and } y > 0$$

$$\log_a x^p = p \log_a x$$

$$\log_a 1 = 0 \text{ and } \log_a a = 1$$

$$\log_a \frac{1}{x} = -\log_a x$$

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